Report on *Matrix Theory Reloaded: A BPS Road to Holography*

Summary. The paper reinterprets the decoupling limits that lead to matrix theories on Dp-branes ("MpT" limits) as BPS limits in which the target space becomes non-Lorentzian. Applied to backreacted D-brane geometries, a single such limit is shown to induce the bulk near-horizon limit of AdS/CFT; iterating limits produces new (including non-Lorentzian-bulk) holographic pairs. Uplifted to M-theory each limit is a DLCQ, motivating the conjecture that holography is a DLCQ^n/DLCQ^m ($m > n$) correspondence. Undoing the limits is identified with $T\bar{T}$-type deformations, giving new p-brane $T\bar{T}$ flow equations for the DBI action.

1. Principal achievement

The main achievement is a unifying, largely algebraic organising principle: the near-horizon limit underlying Maldacena's correspondence is recast as the same BPS decoupling limit that defines matrix theory, so that AdS_5/CFT_4 appears as DLCQ^0/DLCQ^1, BFSS as a special case, and the D1–D5 system as an instance with $m > n + 1$. This gives a systematic generator of holographic dualities (via the table of MpT limits versus Dq-brane stacks) and naturally predicts non-Lorentzian bulk geometries. A concrete and checkable by-product is the family of p-brane generalisations of the $T\bar{T}$ flow equation, obtained by reading off the dependence of brane actions on ω^{-2}; the strikingly simple three-dimensional flow shared by the D2- and M2-brane is particularly appealing. The paper ties together matrix theory, non-Lorentzian geometry and $T\bar{T}$ in one coherent picture.

2. Principal drawback

The central claims rest on classical, kinematic manipulations (scalings of supergravity solutions and of abelian bosonic brane actions) and the DLCQ^n/DLCQ^m statement remains a conjecture with no new dynamical test: no correlator, spectrum, entropy or protected quantity is matched on both sides of any of the novel dualities. Several ingredients are left open by the authors themselves: the non-Lorentzian bulk geometries are obtained by an ad hoc rescaling of ℓ (effectively of N) whose physical meaning is unclear, and it is not verified that they solve the equations of motion and torsional constraints of MpT supergravity (which is not yet formulated); asymptotic regions where the non-Lorentzian structure emerges are where quantum/strong-coupling effects may invalidate the geometric picture (e.g. $p > 3$); and the $T\bar{T}$-like flows are derived only for abelian, bosonic actions, so the claimed link to deforming non-abelian matrix theories into DBI is not established. The paper is therefore strong as a framework but thin as evidence.

3. Most worthwhile follow-up

Balancing feasibility and impact, I would prioritise *establishing the non-Lorentzian bulk geometries as genuine supergravity backgrounds and computing their symmetries*: (i) formulate the bosonic

MpT supergravity equations (e.g. for $p = 0$ or $p = 2$, by expanding type IIA/11d supergravity in ω, as done for non-relativistic M-theory around the M2), (ii) verify that the proposed near-horizon Dq geometries solve them, and (iii) determine their isometry and Killing-spinor superalgebras and compare with the symmetry algebra of the proposed dual (e.g. the M0T super-Bargmann-type algebra and its analogues). This is technically tractable with existing expansion methods, directly addresses the paper's biggest gap, and supplies the first non-trivial symmetry match for a novel "DLCQ^1/DLCQ^2" duality—a prerequisite for any dynamical test (e.g. matching protected spectra or indices, in the spirit of recent BMN finite-N checks). A secondary, lower-risk project is extending the three-dimensional $T\bar{T}$-like flow to supersymmetric and non-abelian (ABJM/BLG) theories.